

NOGOONNUUR, Mongolia

WMO: 442100

Lat: 49.6153N

Lon: 90.2433E

Elev: 2181

StdP: 77.72

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-30.7	-36.1	0.2	-31.6	-34.3	0.2	-29.9	8.8	-10.7	7.4	-10.6	0.6	280	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	27.1	13.1	25.6	12.8	24.3	12.2	14.5	23.6	13.9	22.6	13.2	21.9	3.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.7	11.2	17.2	10.6	10.4	17.6	9.3	9.5	18.0	48.3	23.3	46.4	22.7	44.4	22.0	17.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.5	7.0	-33.7	29.8	3.8	1.7	-36.4	31.0	-38.7	32.0	-40.8	32.9	-43.6	34.2
			-34.2	15.9	3.8	1.1	-36.9	16.7	-39.1	17.3	-41.2	18.0	-44.0	18.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.0	-21.9	-16.5	-4.1	5.1	10.3	17.2	18.7	16.5	9.9	1.9	-9.0	-17.7	(6)
(7)		DBStd	14.60	4.99	5.77	5.94	4.92	4.50	3.30	2.61	3.00	4.40	4.79	5.16	4.48	(7)
(8)		HDD10.0	4112	989	742	437	154	53	1	0	1	54	251	571	858	(8)
(9)		HDD18.3	6419	1248	976	696	397	252	57	27	70	254	508	821	1116	(9)
(10)		CDD10.0	811	0	0	0	8	61	217	269	203	50	2	0	0	(10)
(11)		CDD18.3	77	0	0	0	0	1	23	38	14	1	0	0	0	(11)
(12)		CDH23.3	514	0	0	0	0	13	172	227	93	10	0	0	0	(12)
(13)		CDH26.7	61	0	0	0	0	1	20	30	9	1	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.0	1.2	2.2	2.8	3.5	2.5	2.6	2.2	2.1	2.1	1.7	1.1	(14)
(15)	Precipitation	PrecAvg	166	3	2	5	9	14	29	42	30	12	8	7	5	(15)
(16)		PrecMax	239	9	4	15	18	28	80	108	65	24	16	13	9	(16)
(17)		PrecMin	111	1	1	0	4	4	8	12	8	4	1	2	1	(17)
(18)		PrecStd	34	2	1	4	4	7	15	22	14	6	4	3	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.9	1.0	11.8	19.4	25.2	28.7	29.9	27.9	24.6	16.1	10.0	-1.1	(19)
(20)		MCWB	-8.4	-3.8	3.2	7.5	10.2	13.5	14.5	12.7	11.4	6.1	2.7	-5.0	(20)	
(21)		2%	DB	-8.1	-2.2	9.6	17.4	22.2	26.9	27.2	25.5	21.9	13.4	4.5	-5.1	(21)
(22)		MCWB	-10.8	-6.4	1.5	6.4	9.3	12.8	13.4	13.1	10.4	4.7	-0.4	-7.9	(22)	
(23)		5%	DB	-11.1	-5.0	7.8	15.4	20.4	25.3	25.8	24.1	19.6	11.7	1.8	-8.4	(23)
(24)		MCWB	-13.0	-8.3	0.5	5.4	8.5	12.4	13.1	12.4	9.3	3.9	-2.6	-10.5	(24)	
(25)		10%	DB	-14.1	-7.5	5.8	13.4	18.5	23.8	24.4	22.8	17.6	9.8	-0.8	-11.1	(25)
(26)		MCWB	-15.5	-10.3	-0.7	4.2	7.6	11.6	12.5	11.8	8.3	2.7	-4.5	-12.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.3	-4.0	3.7	8.4	11.2	14.8	16.2	15.2	12.9	6.5	3.7	-5.0	(27)
(28)		MCDWB	-5.6	0.6	10.8	18.0	22.6	25.5	25.2	23.2	21.7	15.0	8.6	-1.2	(28)	
(29)		2%	WB	-10.7	-6.3	1.9	7.1	9.9	14.0	14.7	14.2	11.2	5.2	-0.4	-7.9	(29)
(30)		MCDWB	-8.1	-2.3	8.7	16.3	20.9	23.9	24.0	22.4	20.0	12.7	4.1	-5.1	(30)	
(31)		5%	WB	-13.0	-8.3	0.7	5.7	9.0	13.3	14.1	13.3	9.9	4.1	-2.6	-10.5	(31)
(32)		MCDWB	-11.0	-5.1	7.4	14.6	19.5	22.6	22.6	21.8	18.6	11.3	1.5	-8.6	(32)	
(33)		10%	WB	-15.5	-10.3	-0.7	4.5	8.0	12.7	13.5	12.5	8.8	2.9	-4.4	-12.7	(33)
(34)		MCDWB	-14.1	-7.6	5.5	12.8	17.6	21.7	21.9	21.3	17.1	9.5	-0.9	-11.2	(34)	
(35)	Mean Daily Temperature Range		MDBR	11.0	12.5	12.5	12.0	11.4	10.5	9.4	10.2	11.4	10.8	10.3	9.9	(35)
(36)		5% DB	MCDBR	12.0	13.3	13.1	14.0	13.4	12.2	11.3	11.8	13.4	12.6	12.3	11.4	(36)
(37)			MCWBR	10.1	10.3	8.0	7.5	6.4	4.7	4.6	4.9	6.3	7.5	8.7	9.3	(37)
(38)			MCDBR	12.1	13.5	12.5	12.9	12.2	10.7	9.6	10.7	12.6	12.3	11.8	11.3	(38)
(39)		5% WB	MCWBR	10.2	10.4	7.8	7.3	6.4	4.4	4.1	4.7	6.2	7.5	8.6	9.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.205	0.224	0.259	0.294	0.296	0.302	0.303	0.289	0.258	0.244	0.222	0.205	(40)	
(41)		taud	2.427	2.410	2.384	2.408	2.445	2.496	2.512	2.530	2.574	2.550	2.469	2.412	(41)	
(42)		Ebn at Noon	906	958	965	957	963	956	949	953	960	923	871	852	(42)	
(43)		Edh at Noon	58	77	96	106	107	103	99	93	79	67	55	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.55	2.61	4.09	5.47	6.42	6.55	6.32	5.63	4.50	2.80	1.59	1.20	(44)	
(45)		RadStd	0.06	0.09	0.18	0.21	0.38	0.29	0.35	0.24	0.23	0.18	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	-2.37	N/A	-0.76	-1.11	N/A	N/A	N/A	N/A

Nomenclature: See separate page